

COMPUTER ANIMATION

- ❓ The term **computer animation** generally refers to any time sequence of visual changes in a scene.
- ❓ Animation is the rapid display of a sequence of images to create an illusion of movement. The most common method of presenting animation is as a motion picture or video program.
- ❓ In simpler word; animation means giving life to our imagination.
- ❓ In addition to changing object position with translations or rotations, a computer-generated animation could display time variations in object size, color, transparency, or surface texture.
- ❓ Computer animations can be generated by changing camera parameters, such as position, orientation, and focal length.
- ❓ Computer animations can also be produced by changing lighting effects.

➤ STEPS FOR DESIGN OF ANIMATION SEQUENCES:

AN ANIMATION SEQUENCE IS DESIGNED WITH THE FOLLOWING STEPS:

- ❑ STORYBOARD LAYOUT
- ❑ OBJECT DEFINITIONS
- ❑ KEY-FRAME SPECIFICATIONS
- ❑ GENERATION OF IN-BETWEEN FRAMES(Tweening)

STORYBOARD

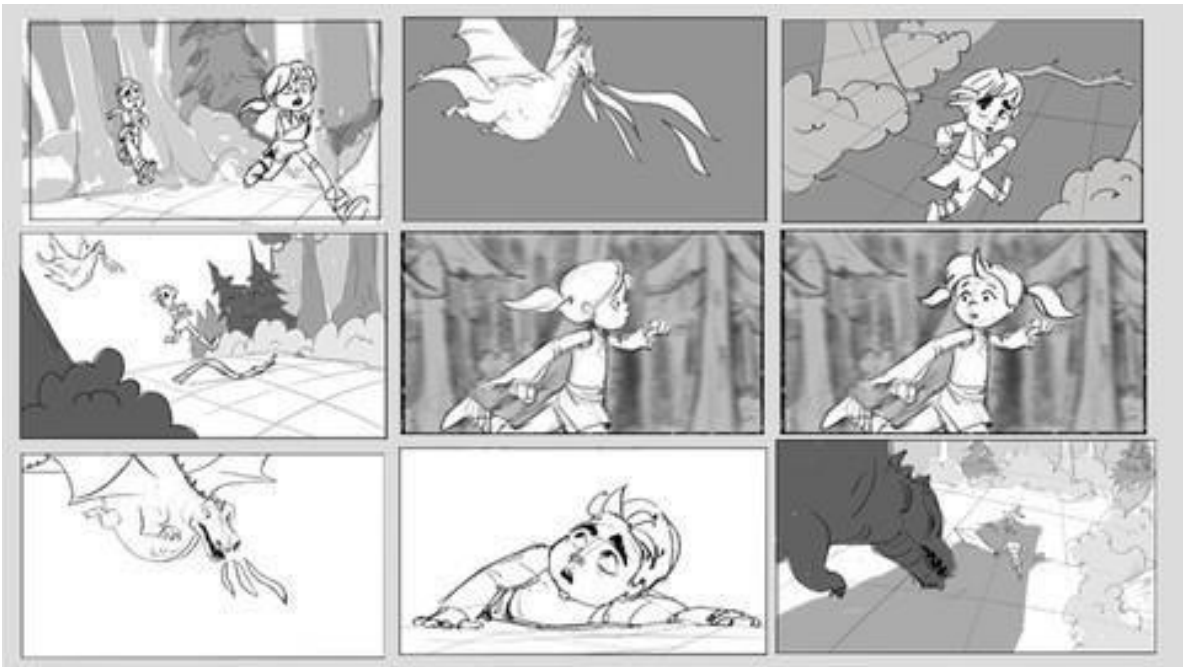
❓ The **storyboard** is an outline of the action. It defines the motion sequence as a set of basic events that are to take place. Depending on the type of animation to be produced, the storyboard could consist of a set of rough sketches or it could be a list of the basic ideas for the motion.

- It acts like a **visual script** of the animation.
- Each panel shows **what happens in a scene**, including:
 - Character actions
 - Camera movements
 - Scene transitions
 - Timing and sequence of events
- Helps the animator plan the flow of the story clearly and systematically.

Importance of Storyboard:

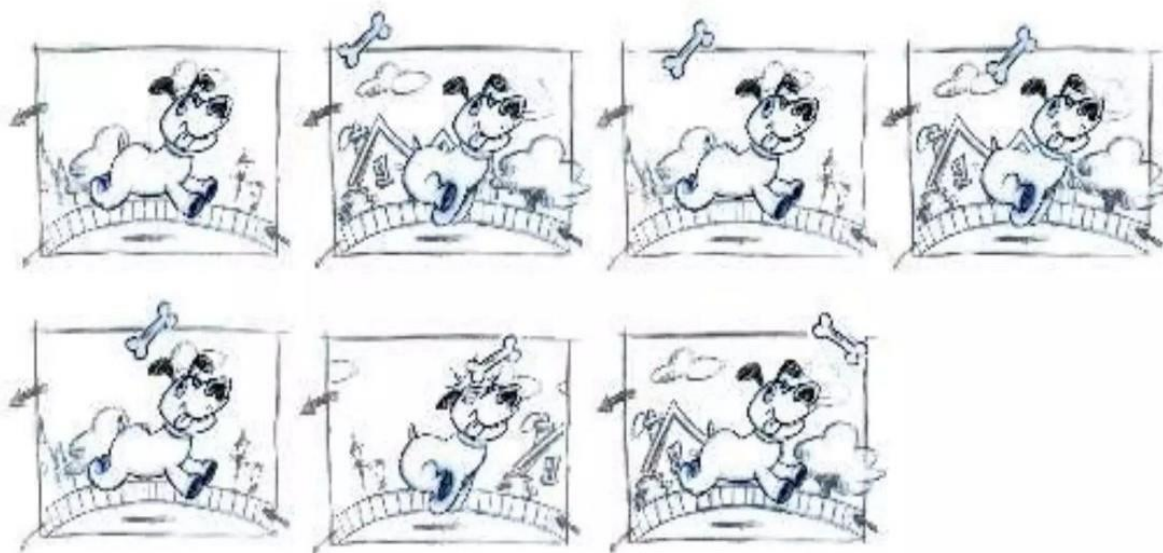
- Provides a **clear blueprint** for the animation.
- Helps identify errors or changes early, saving time and cost.
- Ensures smooth coordination between story, timing, and visuals.

❓



OBJECT DEFINITION

- ❑ An **object definition** is given for each participant in the action. Object can be defined in terms of basic shapes, such as polygons or splines. In addition, the associated movements for each object are specified along with shape. These definitions tell the computer **what the object is and how it should look and behave** during the animation.
- ❑ Objects are defined in keyframes, not in individual frames, because frames are created through tweening in animation.
- ❑
- ❑ • Provides the basic building blocks of the animation.
- ❑ • Ensures consistency of objects across all frames.
- ❑ • Makes it easier to apply transformations and motion.



KEY FRAME

A **key frame** is detailed drawing of the scene at a certain time in the animation sequence. Within each key frame, each object is positioned according to the time for that frame. Some key frames are chosen at extreme positions in the action; others are spaced so that the time interval between key frames is not too great.

?

?

Purpose of Key Frames

?

To define starting and ending points of animation

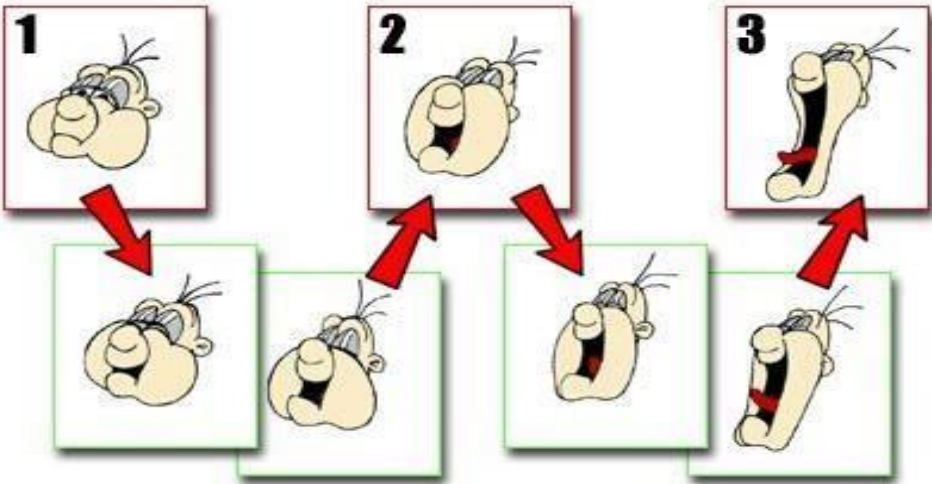
?

To control major changes in objects

?

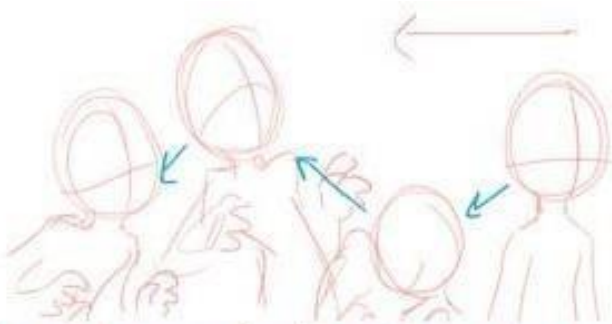
To act as reference points for animation

?

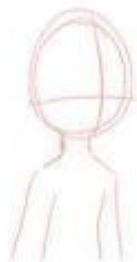


IN-BETWEENS

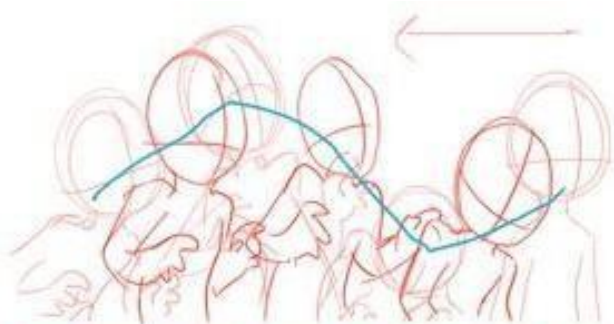
- ❑ **In-betweens** are the intermediate frames between the key frames. The number of in - between needed is determined by the media to be used to display the animation .Film requires 24 frames per second, and graphics terminals are refreshed at the rate of 30 to 60 frames per second . The screen refreshes 60 times per second; the image may remain the same or change based on animation or motion\.
- ❑ A graphics terminal (monitor) does not show an image permanently.
- ❑ It repaints (updates) the screen again and again.
- ❑ Each repaint is called a refresh.
- ❑ The number of times the screen is redrawn in one second is called the refresh rate.



For starters you simply want to create a few frames, a start point and an end point with sometimes a few in between that mark the important motions the character is going to make.



Pretty choppy looking, but we get the basic idea of where and how this character will be moving.



Next you simply go between each pair of frames and draw a pose in between them both so that it leads into the next.



Much better, now we have more movement from the character, but it still looks a little choppy.



Then you simply repeat the process for each pair of frames you add in between to make the animation smoother.



Animation looks a lot better now and has vivid movement.

GENERAL COMPUTER ANIMATION FUNCTIONS:

? **Store and manage the object database.**

? Objects may include shapes, models, characters, backgrounds, etc.

? Each object is stored with its properties such as:

? Shape

? Size

? Color

? Position. The system manages these objects so they can be accessed, modified, or reused during animation.

?

? **Functions for motion generation and for object rendering:** Motion generation controls how objects move over time.

? It defines changes in position, rotation, or size of objects.

? Motion is usually created using key frames and tweening.

? Rendering converts object data into a visual image on the screen.

? It determines how objects appear

? **Functions that simulates camera movements:** Camera simulation defines how the scene is viewed. Common camera movements include:

? Moving forward or backward

? Rotating the view

?

? **Zooming, panning, and tilting:** Makes animation more dynamic and realistic

SOME STEPS INCLUDED THE DEVELOPMENT OF AN ANIM SEQUENCE ARE:

- ☐ Object manipulations and rendering
- ☐ Camera motions
- ☐ Generation of in-betweens

-
-
-
- ❏ Animation packages, *such as Wave-front*, provide special functions for designing the animation and processing individual objects.
 - ❏ In animation packages, one function is provided *to store and manage the object database*. Object shapes are stored and updated in the database.
 - ❏ Other object functions include those *for motion generation and object rendering*.
 - ❏ Motions can be generated according to specified constraints using two dimensional or three-dimensional transformations.
 - ❏ Another function motions *simulates* *camera* *movements*. Standard zooming, panning, and tilting.

COMPUTER ANIMATION LANGUAGES:



1. C
2. Lisp
3. Pascal
4. FORTRAN
5. PYTHON
6. C#
7. Special animation packages like Adobe Flash , autodesk maya etc.

- ❓ Design and control of animation sequences are handled with a set of animation outlines.
- ❓ A general purpose language, such as **C, python,c#,Lisp, Pascal or FORTRAN** is often used to program the animation functions, but several **specialized animation languages** have been developed.
- ❓ Animation functions include a **graphics editor, a key frame generator, an in-between generator and standard graphics routines.**
- ❓ The graphics editor allows us to design and modify object shapes, using spline surface, constructive solid geometry methods, or other representation schemes.

RASTER ANIMATIONS:

- ❑ Raster animation is a type of animation where images are created using pixels arranged in a grid (bitmap images), and motion is produced by displaying a sequence of raster images quickly one after another.
- ❑ Raster animation works by showing multiple bitmap images (frames) in rapid succession. When these frames change slightly from one to another, the human eye perceives it as movement.
- ❑ Example process:
 - ❑ Create the first image (frame 1).
 - ❑ Slightly modify the object position (frame 2).
 - ❑ Continue creating many frames.
 - ❑ Display them quickly (e.g., 24 frames per second).
 - ❑ This creates the illusion of motion.
- ❑ **Advantages**
 - ❑ Good for detailed images and realistic graphics
 - ❑ Widely supported by many applications
 - ❑ Easy to edit using image editors
- ❑ **Disadvantages**
 - ❑ Large file sizes
 - ❑ Quality reduces when scaled
 - ❑ Requires many images for animation
- ❑ On raster scan systems we can generate real time animation in limited applications using **raster operations**.
- ❑ A simple method for translation in the xy plane is to transfer a rectangular blocks of pixels through arbitrary angles using anti-aliasing procedures.
- ❑ To rotate a block of pixels we need to determine the percent of area coverage for those pixels that overlap the rotated block.
- ❑ Sequences of raster operations can be executed to produce real time animation of either two-dimensional or three-dimensional objects, as long as we restrict the animation to motions in the projection plane.
- ❑ Then no viewing or visible surface algorithms need be invoked.
- ❑ **Color table transformations** also produce animation. Just by changing the intensity values of pixels to on and off, we can produce animation on raster systems.

➤ Motion Specification In Computer Animation

There are several ways in which the motions of objects can be specified in an animation system. We can define motion in very explicit terms, or We can use more abstract or more general approaches.



➤ Direct motion specification : -

The most straightforward method for defining a motion sequence is direct specification of the motion parameters. Here, We explicitly give the rotation angles and translation vectors. Then the geometric transformation matrices are applied to transform co-ordinate positions. Alternatively, We could use an approximating equation to specify certain kinds of motions. These methods can be used for simple user programmed animation sequences.

➤ **Goal-directed systems : -**

At the opposite extreme, We can specify the motions that are to take place in general terms that abstractly describe the actions. these systems are referred to as goal directed because they determine specific motion parameters given the goals of the animation. For example, We could specify that we want an object to "walk " or to "run" to a particular destination. Or We could state that we want an object to "pick up " some other specified object. The input directive are then interpreted in term of component motions that will accomplish the selected task. Human motion, for instance, can be defined as a heirarchical structure of sub motion for the toros, limbs,and so forth.

➤ **Kinematics and dynemics : -**

We can also construct animation sequences using kinematic or dynemic descriptions. With a kinematic description, we specify the animation by giving motion parameters position, velocity, and acceleration) without reference to the forces that cause the motion. for constant velocity (zero acceleration), we designate the motions of rigid bodies in a scene by giving an initial position and velocity vector for each objects.

An alternate apporach is to use inverse kinematics. Here, we specify the initial and final positions of objects at specified times and the motion parameters are computed by the system . For example, assuming zero acceleration , we can determine the constant velocity that will accomplish the movement of an object from the initial position to the final position.

Dynamic descriptions on the other hand, require the specification of the forces that produce the velocities and acceleration. Descriptions of object behavior under the are generally referred to as a physically based modeling. Example of forces affecting object motion include electromagnetic, gravitational, friction, and other mechanical

forces.

Object motion are obtained from the forces equations describing physical laws, such as newton's law of motion for gravitational and friction processes, euler or navier-stokes equations describing fluid flow, and maxwell 's equations for electromagnetic forces. For example, the general form of newton's second law for a particle of mass m is

$$\mathbf{F} = d(m\mathbf{v})/dt$$

with $\mathbf{F}$ as the force vector, and $\mathbf{v}$ as the velocity vector. If mass is constant, we solve the equation $\mathbf{F} = m\mathbf{a}$, where $\mathbf{a}$ is the acceleration vector. otherwise, mass is a function of time, as in relativistic

motions of space vehicles that consume measurable amounts of fuel per unit time. We can also use inverse dynamics to obtain the forces, given the initial and final positions of objects and the type of motion.

Application of physically based modeling include complex rigid-body systems and such nonrigid systems as cloth and plastic materials. Typically, numerical methods are used to obtain the motion parameters incrementally from the dynamical equations using initial conditions or boundary values.

➤ KEY-FRAME SYSTEMS

What is a Keyframe?

A **keyframe** is a drawing or frame that defines the **starting and ending points** of a movement in animation. It shows the **important positions** of an object at specific times.

- Frames are measured in **units of time** (frames on film or video).
- A sequence of keyframes determines **what movement** the viewer sees.
- The **distance between keyframes** decides the **speed and timing** of the animation.

2. In-Between Frames

Only two or three keyframes per second cannot create smooth motion.

- The missing frames are filled with **in-between frames (tweens)**.
- In-betweens create the **illusion of smooth movement** between keyframes.

3. Keyframes in Complex Animation

In complex animations, object shapes may change over time, such as:

- Facial expressions
- Clothes movement
- Exploding or disintegrating objects
- Transforming one object into another

If objects are created using **polygon meshes**, the number of edges and lines may change in different frames. So, each keyframe may have **different geometry**.

4. Keyframe Characteristics (Adobe After Effects)

- ✔ **Multiple keyframes** can exist on a single layer.
- ✔ Keyframes can be added to:
 - Text
 - Shapes
 - Images
 - Lines
- You cannot add keyframes to:
 - A locked element
 - A normal layer as a whole (unless it is a compound layer)

Order of Keyframes

- The **first keyframe** added has priority.
- Additional keyframes are interpreted based on the first one.
- If no change is applied, objects behave as **independent elements**.

5. Origin of Keyframes

- Keyframes are the **first points in time** used to create animation.
- Animation **cannot start without keyframes**.
- The origin of keyframes is the same as frames—they define **time and motion**.

6. Keyframes in Flash (CS5 and Earlier)

- When a keyframe is created, it is set to **0% by default**.

-
-
-
- Moving the timeline forward may show **no movement** unless keyframes are adjusted.
 - This limitation can be solved using **ActionScript (AS4 or later)**.
 - Earlier Flash versions do not support this solution.

7. Layout and Keyframing in Animation

Animation is an **art**, and keyframing is the most common technique.

Keyframing means:

- Creating new frames based on previous frames
- Adding or removing frames to control movement and timing

Example:

- You have 10 frames showing an action.
- You add extra frames to create **pauses**.
- Blank or repeated frames slow down motion and improve **storytelling**.

8. Keyframes in 2D and 3D Animation

2D Animation

- A keyframe is created by changing:
 - Position
 - Rotation
 - Scale of an object

3D Animation

- Keyframes are created using the **Timeline Editor**.
- It allows:
 - Forward and backward time movement
 - Creation of in-between frames
 - Adding effects like fades, transitions, and sound

9. Advantages of Keyframing

- ✓ Smooth animation
- ✓ Better timing control
- ✓ Easier creation of complex scenes
- ✓ Multiple layers can be animated together
- ✓ No need to manually draw every frame



A key-frame in animation is a drawing that defines the starting and ending points of any smooth transition.

The drawings are called "frames" because their position in time is measured in frames on a strip of film. A

sequence of key frames defines which movement the viewer will see, whereas the position of the key frames

on the film, video or animation defines the timing of the movement. Because only two or three key frames

over the span of a second do not create the illusion of movement, the remaining frames are filled with in

betweens. With complex object transformations, the shapes of objects may change over time. Examples are

clothes, facial features, magnified detail, evolving shapes, exploding or disintegrating objects, and

transforming one object into another object. If all surfaces are described with polygon meshes, then the

number of edges per polygon can change from one frame to the next.

Thus, the total number of line segments can be different in different frames. Keyframe Characteristics Here

are some of the keyframe characteristics that you'll need to be aware of when animating in Adobe After

Effects:-You can have multiple keyframes on a single layer. For example, if you have an object moving



across the screen at different speeds, you can create two separate layers and then animate the first one independently of the second.-You can add keyframes to any element within a layer. This includes text, shapes, lines and even pictures. However, you cannot add keyframes to an entire layer (unless it is a compound layer). You also cannot add keyframes to an element that is already locked down before adding new ones.

The order in which you add new keyframes affects how they are interpreted by After Effects. The first one added will take priority over subsequent ones once it has been added. In other words, if you add a second set of keyframes but don't change anything else about them (e.g., rotation), then both sets will be rendered as if they were fully independent objects. The Origin Of Keyframes The origin of keyframes is the same as that of frames. Keyframes are the first points in time for animation, so they must be set up before you can begin animating. The origin of keyframes is the same as that of frames.

Keyframes are the first points in time for animation, so they must be set up before you can begin animating. In Flash CS5 and earlier versions, when you create a keyframe for an object's position or rotation, it's automatically set to 0%. This means that if you move your timeline indicator along the timeline to a point past where the object should be at that frame, you'll see nothing happen because it's not there yet! One way to

overcome this problem is to use ActionScript to create a script that sets your timeline indicator position back to 0% whenever it reaches the last frame on your timeline (the first frame after which no action will take place).

However, this method only works if you're using ActionScript 4 or later; in earlier versions of Flash CS5 and below, it won't work because without AS 4 installed you can't access any variables from within ActionScript 3 scripts. How To Layout And Keyframe In Animation Animation is an art form, and like all forms of art, it's difficult to master. There are many ways to create animations, but the most common way is to keyframe them. Keyframing refers to the act of creating a new animation frame based on the previous one. In other words, you can add or remove frames from your animation based on what you want the audience to see. The most basic form of keyframing involves adding or removing some number of frames in between two existing frames in your sequence.

For example: 1) You have a sequence of 10 frames that you want your audience to see as a single frame (the final frame). 2) You want to add 3 more frames where there should be a pause at the end of each line or section (such as when one character says something and another responds). So you make 10 frames and add 3 more as follows: 3) The first 8 are exactly the same as before (except for adding those three extra lines), but then there's a pause at the end of each line/section so you insert two blank shots after each line/section where there should be no characters present (so they're out of shot).

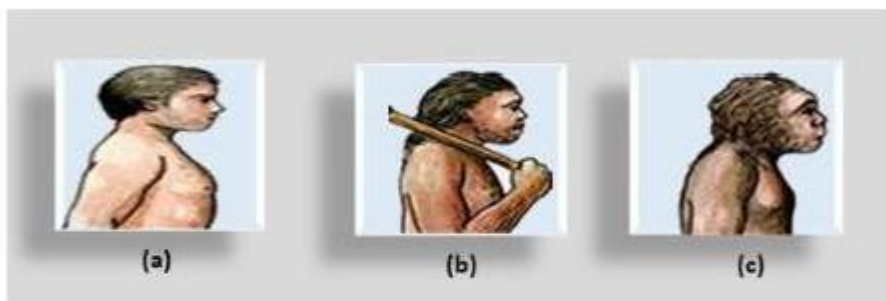
Keyframes are the points in an animation's timeline at which a change occurs. They're also called control points, or just "key." In traditional 2D animation, a keyframe is created by moving an object's position or rotation to a new location on the stage. In 3D animation, it's easy to create and edit keyframes using the Timeline Editor. The Timeline Editor allows you to select a point in time and move it forward or backward (with respect to time) so that you can set up your storyboard. The editor also lets you create "in-between" frames those intermediary positions between two other frames and add special effects such as fades between scenes or special transitions between characters' speech bubbles or Leave a Reply Your comment * even sound effects. You can even animate with multiple layers at once, making it possible for you to create complex animations without having to manually track each individual frame.

➤ Animation Functions

1. Morphing: Morphing is an animation function which is used to transform object shape from one form to another is called Morphing. It is one of the most complicated transformations. This function is commonly used in movies, cartoons, advertisement, and computer games.

For Example:

1. Human Face is converted into animal face as shown in fig:



2. Face of Young person is converted into aged person as shown in fig:



The process of Morphing involves three steps:

1. In the first step, one initial image and other final image are added to morphing application as shown in fig:
1st & 4th object consider as key frames.
2. The second step involves the selection of key points on both the images for a smooth transition between two images as shown in 2nd object.

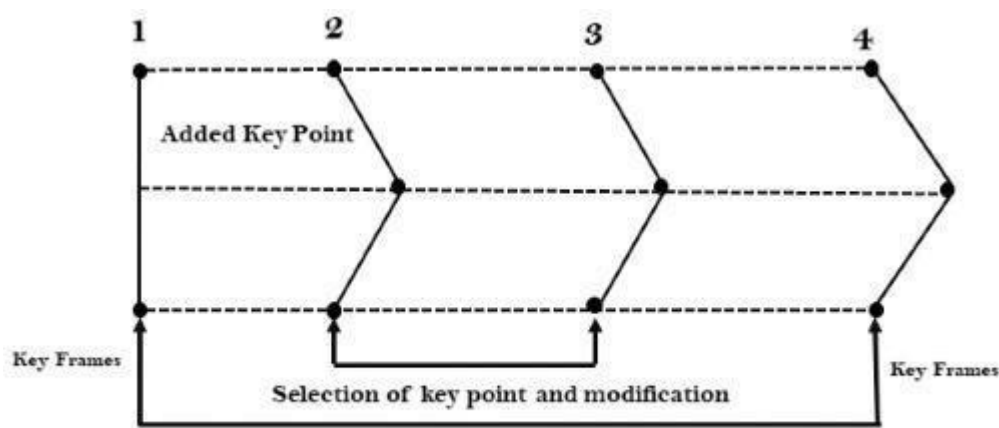


Fig: Process of Morphing

3. In the third step, the key point of the first image transforms to a corresponding key point of the second image as shown in 3rd object of the figure.

2. Select the **initial image** and the **final image**
3. These act as **key frames**
4. Both images should have:
 1. Similar pose
 2. Similar orientation
 3. Similar scale (if possible)

Place Control Points Manually

- Use the morphing tool to **click and place points**
- Place points in:
 - Same **order**
 - Same **relative position**
- Each point in image 1 must have a **matching point** in image 2

Example:

- Left eye corner in image 1 → Left eye corner in image 2

control points are implicitly defined as mesh vertices. The user manually selects and modifies corresponding vertices on the 3D mesh to achieve morphing using blend shapes.

5. **Wrapping:** Wrapping function is similar to morphing function. It distorts only the initial images so that it matches with final images and no fade occurs in this function.

6. **Tweening:** Tweening is the short form of 'inbetweening.' Tweening is the process of generating intermediate frames between the initial & last final images. This function is popular in the film industry.

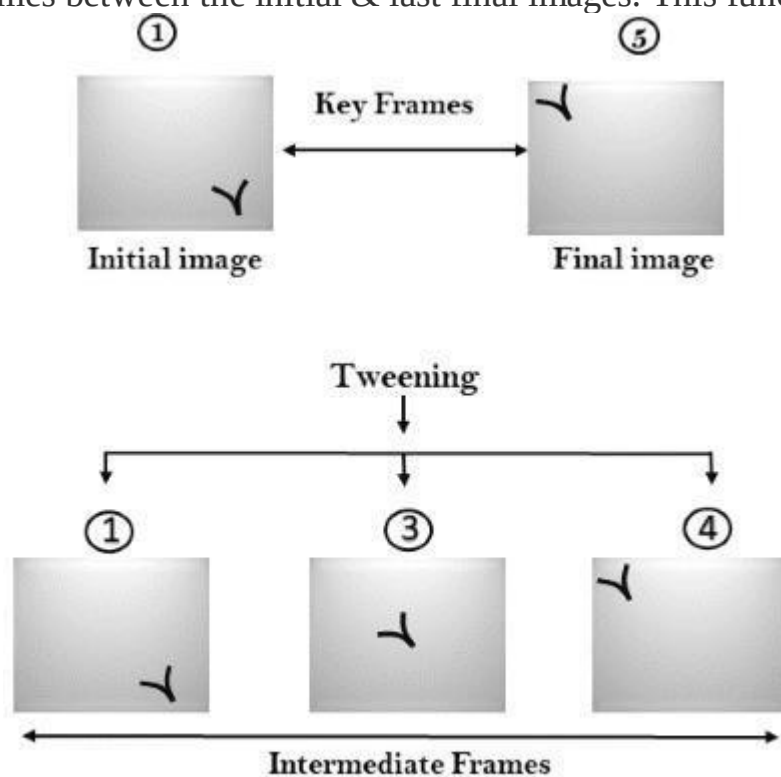


Fig: Tweening

7. **Panning:** Usually Panning refers to rotation of the camera in horizontal Plane. In computer graphics, Panning relates to the movement of fixed size window across the

window object in a scene. In which direction the fixed sized window moves, the object appears to move in the opposite direction as shown in fig:

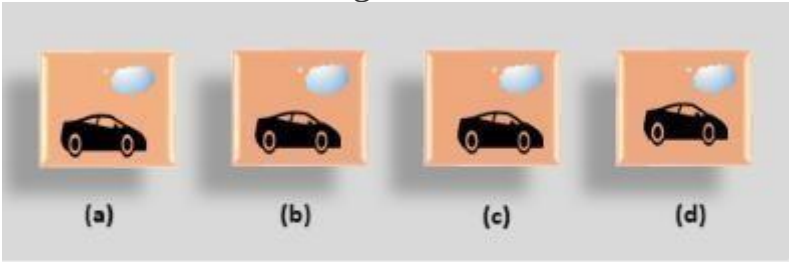


Fig: Panning

If the window moves in a backward direction, then the object appear to move in the forward direction and the window moves in forward direction then the object appear to move in a backward direction.

8. **Zooming:** In zooming, the window is fixed an object and change its size, the object also appear to change in size. When the window is made smaller about a fixed center, the object comes inside the window appear more enlarged. This feature is known as **Zooming In**. When we increase the size of the window about the fixed center, the object comes inside the window appear small. This feature is known as **Zooming Out**.

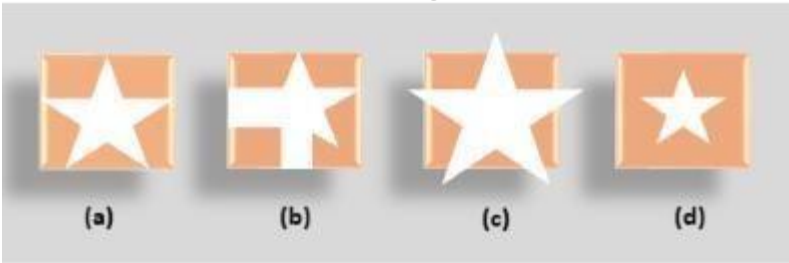


Fig: Zooming in & Zooming Out

9. **Fractals:** Fractal Function is used to generate a complex picture by using Iteration. Iteration means the repetition of a single formula again & again with slightly different value based on the previous iteration result. These results are displayed on the screen in the form of the display picture.